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AREA

Counting Squares & Formula

Class 2 • Mathematics

A complete lesson covering concepts, methods, formulas, examples, practice problems, and real-life connections.

■ Topics Covered	■ Duration	■ Learning Goals
• What is Area? • Unit Squares • Counting Method • Formula ($L \times W$) • Area vs Perimeter	45 – 60 min (4–5 sections)	• Define area correctly • Count squares accurately • Apply $\text{Area} = L \times W$ • Distinguish from perimeter

Section 1: What is Area?

Area is the amount of **flat space** a shape covers. We measure area to understand how big or small the inside of a shape is.



Key Definition

Area = The flat space **INSIDE** a shape
Bigger shape → Bigger area
Smaller shape → Smaller area

■ The Chapati Challenge ■

Dadi is making chapatis. She places TWO chapatis on the tava — one BIG and one SMALL.

Question: Which chapati covers MORE space?

Answer: The BIG chapati — because it has a LARGER AREA!

This is exactly what area means: how much surface a flat object covers.

Real-Life Connections

■ Home	■ School	■ Garden	■ Festival
Mum buys floor tiles — she needs the area of the kitchen.	Priya decorates her notebook — area of cover for sticker paper.	Mali plants grass — area of the school ground.	Rangoli design — area of floor covered during Diwali.

Section 2: Unit Squares

A **unit square** is a square that is **1 unit long** and **1 unit wide**. It is the basic building block for measuring area.

Area of one unit square:

1 sq cm (also written as 1 cm²)

Think of it like this: One small tile on your kitchen floor is exactly ONE unit square. Count those tiles → that's the area!



1 unit square = 1 sq cm



6 unit squares = 6 sq cm



12 unit squares = 12 sq cm



Important Units

Always write the unit after your answer!

- sq cm or cm² (for small shapes)
- sq m or m² (for large shapes like rooms)

Without the unit, your answer is INCOMPLETE.

Section 3: Counting Squares Method

The **Counting Squares Method** is the simplest way to find area. We place a shape on a grid and count how many unit squares fit inside it.

Step-by-Step How to Count:

- 1 Place the shape on squared (grid) paper.
- 2 Count every complete square inside the shape.
- 3 Write the total number, then add the unit (sq cm).
- 4 Double-check by counting row by row.

Worked Example 1: Rectangle

1	2	3	4
5	6	7	8
9	10	11	12

Rectangle: 3 rows $\times$ 4 columns

Count the squares:

- Row 1: 4 squares
- Row 2: 4 squares
- Row 3: 4 squares

Total = $4 + 4 + 4 = 12$ squares

Area = 12 sq cm ✓

Worked Example 2: L-Shape

1	2	3
4		
5		
6	7	8

L-Shape: 8 unit squares

Count each square carefully:

Top row: 3 squares (columns 1–3)

Left side: 5 squares (rows 2–4, column 1) → but shared = actually count...

Let's count all: 1, 2, 3, 4, 5, 6, 7, 8 squares

Area = 8 sq cm ✓

Same Area, Different Shapes!

A rectangle (2 rows $\times$ 5 cols) has area = 10 sq cm.

An L-shape can also have area = 10 sq cm.

Two shapes can look COMPLETELY different but have the SAME area!

Area is about the INSIDE SPACE, not the shape's outline.

Section 4: The Formula — Length × Width

Counting squares works well for small shapes, but what if the shape has 100 squares? We need a **shortcut!**

Look at a rectangle carefully. Notice that the squares are arranged in neat **rows** and **columns**:

Rectangle:

Also: Area = Rows × Columns

Area = Length × Width

Why does it work?

Step	What we do	Result
1	Count squares in 1 row	= Width (4 squares)
2	Count how many rows	= Length (3 rows)
3	Multiply (instead of adding the row 3 times)	3 × 4 = 12 sq cm
✓	Area = Length × Width	12 sq cm

Formula for a SQUARE

All sides of a square are **equal**. So we multiply the same number twice:

Square (all sides equal):

Example: 3 cm × 3 cm = 9 sq cm

Area = Side × Side

Shape	Formula	Example	Area
Rectangle	L × W	6 m × 4 m	24 sq m
Square	S × S	5 cm × 5 cm	25 sq cm
Irregular / L-shape	Count squares	Count = 8	8 sq cm

Section 5: Area vs Perimeter

Students often confuse **area** and **perimeter**. They are different measurements that answer different questions.

■ AREA	■ PERIMETER
The space INSIDE a shape	The distance AROUND a shape
Measured in square units (sq cm, m ²)	Measured in units (cm, m)
Think: grass INSIDE the garden	Think: fence AROUND the garden
Formula: $L \times W$	Formula: add all sides
Rectangle $5\text{m} \times 3\text{m} \rightarrow \text{Area} = 15 \text{ sq m}$	Rectangle $5\text{m} \times 3\text{m} \rightarrow \text{Perimeter} = 5+3+5+3 = 16 \text{ m}$

■ **Never Mix Them Up!**

Area answers: 'How much space inside?' $\rightarrow 15 \text{ sq m}$
Perimeter answers: 'How far around?' $\rightarrow 16 \text{ m}$
Different values, different units — always be careful!

Section 6: Key Vocabulary

Word	Meaning	Example
Area	The flat space INSIDE a shape	Area of a tile
Unit Square	A $1\text{ cm} \times 1\text{ cm}$ square — our basic unit	One grid square
Square Unit	The unit we use for area (sq cm, cm^2 , sq m)	12 sq cm
Length	How long a shape is (longer side)	5 cm
Width	How wide a shape is (shorter side)	3 cm
Perimeter	The distance all the way around a shape	16 m
Row	A horizontal line of squares (left → right)	Row 1, Row 2...
Column	A vertical line of squares (top → bottom)	Col 1, Col 2...

Section 7: Practice Problems

Solve each problem. Show your working. Don't forget the units!

Q1 — Count the Squares

A shape on a grid has squares arranged in 2 rows and 6 columns.
How many unit squares are inside? What is the area?

Answer: Counting: $2 \times 6 = 12 \rightarrow \text{Area} = 12 \text{ sq cm}$

Q2 — Rectangle Formula

A classroom floor is 8 m long and 5 m wide.
Find the area of the floor using the formula.

Answer: Area = Length $\times$ Width = $8 \times 5 = 40 \text{ sq m}$

Q3 — Square Formula

A window pane is 3 m on each side.
Calculate the area of the window.

Answer: Area = Side $\times$ Side = $3 \times 3 = 9 \text{ sq m}$

Q4 — Compare Shapes

Shape A: Rectangle — $4 \text{ cm} \times 3 \text{ cm}$
Shape B: Rectangle — $6 \text{ cm} \times 2 \text{ cm}$
Which shape has the larger area? Are they equal?

Answer: A: $4 \times 3 = 12 \text{ sq cm}$ B: $6 \times 2 = 12 \text{ sq cm} \rightarrow \text{Both EQUAL! } 12 \text{ sq cm}$

Q5 — L-Shape

An L-shaped room has unit squares counted as:
Top row: 5 squares | Left column: 3 extra squares.
What is the total area?

Answer: $5 + 3 = 8 \text{ sq cm}$

Q6 — Area vs Perimeter

A garden is 7 m long and 4 m wide.
(a) Find the area. (b) Find the perimeter.

**Answer: (a) Area = $7 \times 4 = 28 \text{ sq m}$
(b) Perimeter = $7 + 4 + 7 + 4 = 22 \text{ m}$**

Section 8: Exit Ticket Quiz

Instructions: Answer all questions. Write clearly. Show all working.

#	Question	My Answer	Marks
1	A grid shape has 8 unit squares inside it. What is the area?		1
2	A rectangle is 5 cm long and 3 cm wide. Use the formula to find the area.		2
3	True or False: 'Area and perimeter are the same thing.'		1
4	A square has a side of 4 cm. What is its area?		2
5	Which has more area? Shape A: 3×6 cm OR Shape B: 4×5 cm		2

Answer Key (Teacher Reference):

Q1	Q2	Q3	Q4	Q5
8 sq cm	15 sq cm	FALSE	16 sq cm	Shape B: 20 sq cm (A=18 sq cm)

Section 9: Lesson Summary

✓ 3 Ways to Find Area

1. COUNT the unit squares inside the shape (grid method)
2. USE THE FORMULA: $\text{Area} = \text{Length} \times \text{Width}$ (for rectangles & squares)
3. ALWAYS write the UNIT (sq cm or cm^2) after your number!

■ Shabash! You are an AREA EXPERT! ■

You have learned:

- ★ What area means
- ★ How to count unit squares
- ★ The formula $\text{Length} \times \text{Width}$
- ★ Area vs Perimeter
- ★ Real-life uses of area

Keep practising — area is all around you! ■